AMC ENGINEERING COLLEGE

DEPARTMENT OF PHYSICS

Module 4 Notes

Electrical Properties of Materials and Applications

I/II SEMESTER Physics for CSE Stream

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Contents

1 Electrical Conductivity in Metals **5**

1.1 Introduction . . . 5
1.2 Failures of Classical Free Electron Theory . . . 6
1.2.1 Specific Heat . . . 7
1.2.2 Dependance of Electrical Conductivity on Temperature . . . 7
1.2.3 Dependance of Electrical Conductivity on Electron Concentration . . . 7
1.3 Concept of Phonon . . . 8
1.4 Mattheissen's rule . . . 8
1.5 Quantum Free Electron Theory . . . 9
1.5.1 Assumptions of Quantum Free Electron Theory . . . 9
1.5.2 Density of States . . . 9
1.5.3 Fermi Energy . . . 10
1.6 Fermi-Dirac Statistics . . . 10
1.6.1 Dependence of Fermi Factor on Temperature and Energy . . . 10
1.6.2 Worked Examples/Numericals . . . 11
1.7 Model and Previous Year Questions . . . 13

2 Superconductivity **15**

2.1 Introduction . . . 15
2.1.1 Definition . . . 16
2.1.2 Critical Temperature . . . 16
2.1.3 Variation of resistivity with temperature . . . 16
2.2 Meissner's Effect . . . 16
2.2.1 Critical Field . . . 19
2.2.2 Worked Examples . . . 20
2.3 Types of Superconductors . . . 21
2.3.1 Type I superconductors . . . 21
2.3.2 Type II superconductors . . . 21
2.4 BCS Theory of Superconductivity . . . 22
2.5 High-temperature Superconductors . . . 23
2.6 Quantum Tunnelling . . . 24
2.7 Josephson Junction . . . 25
2.7.1 DC Josephson Effect . . . 25
2.7.2 AC Josephson Effect . . . 26
2.8 SQUID . . . 26
2.8.1 Flux Quantization . . . 26
2.8.2 DC SQUID . . . 27
2.8.3 RF SQUID . . . 27
2.9 Applications of superconductors in quantum computing . . . 28

2.9.1 Charge qubit 28
2.9.2 Flux qubit 28
2.9.3 Phase qubit 29
2.10 Model and Previous Year Questions 29
2.11 Numericals 29

Chapter 1

Electrical Conductivity in Metals

Syllabus
Electrical Conductivity in Metals
Resistivity and Mobility, Concept of Phonon, Matheissen's rule, Failures of Classical Free Electron Theory, Assumptions of Quantum Free Electron Theory, Fermi Energy, Density of States, Fermi Factor, Variation of Fermi Factor With Temperature and Energy. Numerical Problems.

1.1 Introduction

Metallic solids have atoms bound together by metallic bonds. In such solids, the valence electrons are free to move about in the whole body of the solid and hence contribute to conduction in it. Such electrons do not belong to a particular atom but belong to the material as a whole. Normally, one or two electrons per atom are free electrons and they move about randomly in the absence of electric field. This motion is identical with the motion of molecules in a gas, and hence it is called electron gas. This electron gas is responsible for high electrical and thermal conductivity in metals.

Classical Free Electron Theory of Metals (Only for reference)

Drude and Lorentz developed the classical free electron theory based on the following assumptions:

- Metal atoms have some of the valence electrons freely moving about in the body of the metal. These electrons are called free electrons or conduction electrons as they contribute to conduction in the metal.
- The electrostatic force of attraction between the free electron and the ion cores is negligible. Also, the electrostatic force of repulsion amongst the free electrons is negligible.
- Due to the above assumption, we can say that the free electrons are equivalent to molecules in a gas and thus are assumed to obey the laws of kinetic theory of gases. In the absence of external electric field, they are at random motion and their average kinetic energy is given by $\frac{1}{2}mv_{th}^2 = \frac{3}{2}k_BT$, where v_{th} is the average thermal velocity of the electrons, m is the mass of the electron, k_B is the Boltzmann's constant, and T is the temperature.

- Inside the metal, the electric potential due to the ionic cores is constant and the electrons cannot escape from the metal due to the potential barrier at the surface.

Terminologies

Thermal Velocity (v_{th}): The average velocity of random motion of free electrons within the body of the metal in the absence of external field is called thermal velocity.

Mean Free Path (λ): It is the average distance travelled by the free electrons between successive collisions with the lattice ions.

Mean collision time (τ): The average time interval between successive collisions of the free electrons with the lattice ions is called mean collision time. If v is the average velocity of electrons and λ is the mean free path, then $\tau = \lambda/v$.

Drift velocity (v_d): When an electric field is applied across the metal, some of the free electrons acquire a constant, terminal velocity in a direction opposite to that of the field. This steady and very small velocity is called drift velocity.

Current Density (J): It is the electric current (I) passing normally through unit area of cross-section (A) of the conductor, i.e., $J = I/A$. But we know that $I = nAev_d$. Therefore, $J = nev_d$, where n is number density of electrons, e is charge of electrons, v_d is drift velocity of electrons. Unit of current density is Am^{-2}.

Electric Field (E): It is the potential difference (V) across unit length (L) of a conductor, i.e., $E = V/L$. Unit of electric filed is Vm^{-1}.

Conductivity (σ): It is the quantity that characterizes the ability of a material to conduct electricity. It is the reciprocal of resistivity (ρ), i.e., $\sigma = 1/\rho$. (**Resistivity** is the property of the material by virtue of which the flow of current through it is opposed.) As per the classical free electron theory, we have

$$\sigma = \frac{ne^2\tau}{m}$$

where n is the free electron concentration, e is charge of electron, m is mass of electron and τ is the mean collision time.

Mobility (μ): It is the magnitude of drift velocity (v_d) acquired per unit electric field (E), i.e.,

$$\mu = \frac{v_d}{E}$$

But $v_d = \frac{eE\tau}{m}$. So we have

$$\mu = \frac{e\tau}{m}$$

1.2 Failures of Classical Free Electron Theory

The classical free electron theory could successfully explain electrical and thermal conduction in metals. But it could not account for specific heat of metals, temperature and free electron concentration dependence of electrical conductivity.

1.2.1 Specific Heat

According to the classical theory, the molar heat capacity of free electrons in a metal is $\frac{3}{2}R$. However, the experimentally determined molar heat capacity of metals is $10^{-4}RT$, which is very much lower than the theoretically predicted value and also shows a dependence on temperature which is not predicted by the classical free electron theory.

1.2.2 Dependance of Electrical Conductivity on Temperature

According to the assumption of classical free electron theory, we have

$$\frac{1}{2}mv_{th}^2 = \frac{3}{2}k_BT \implies v_{th} \propto \sqrt{T}$$

Since mean collision time is inversely proportional to thermal velocity, we can write

$$\tau \propto \frac{1}{v_{th}} \implies \tau \propto \frac{1}{\sqrt{T}}$$

But we know that

$$\sigma = \frac{ne^2\tau}{m}$$
$$\sigma \propto \tau$$
$$\implies \sigma \propto \frac{1}{\sqrt{T}}$$

So from theory, we see that conductivity is inversely proportional to the square root of temperature. But experimentally, it was found that conductivity was inversely proportional to temperature, i.e., $\sigma \propto \frac{1}{T}$. Thus the classical theory couldn't explain the temperature dependance of conductivity.

1.2.3 Dependance of Electrical Conductivity on Electron Concentration

From the expression for electrical conductivity $\sigma = \frac{ne^2\tau}{m}$, conductivity is directly proportional to the electron density n. Hence, divalent and trivalent metals should possess much higher electrical conductivity than monovalent metals. This is contrary to the experimental observation that silver and copper (monovalent metals) are more conducting than zinc and aluminium (divalent and trivalent metals). The experimental values of conductivity of some metals are listed below:

Type	**Metals**	**Conductivity** σ (Sm^{-1})	**Electron Concentration** n (cm^{-3})
Monovalent	Copper	5.88×10^7	8.4×10^{28}
	Silver	4.5×10^7	5.8×10^{28}
Divalent	Zinc	1.09×10^7	13×10^{28}
	Cadmium	0.15×10^7	9.2×10^{28}
Trivalent	Aluminium	3.65×10^7	18.06×10^{28}
	Silver	0.67×10^7	15.3×10^{28}

From the above values it is clear that the conductivity is low for metals with high electron concentration and conductivity high for those metals with lower electron concentration. Thus classical free electron theory could not explain the dependence of conductivity on free electron concentration.

1.3 Concept of Phonon

In the free electron theory, electrons in a crystal are treated as if they move independently, almost as if they were free particles in a metal. However, in reality, the crystal lattice, which consists of atoms or ions held in a regular repeating pattern, is not completely still. The atoms or ions in the lattice are in constant motion, vibrating about their equilibrium positions due to thermal energy. These vibrations give rise to what we call as phonons. *Phonons represent the vibrational modes of atoms or ions in a crystal lattice.*

Phonons are quantized units of lattice vibrations, similar to how photons are quantized units of light. They carry both energy and momentum. When an electron moves through the crystal lattice, it can interact with phonons. These interactions can scatter electrons, leading to phenomena like electrical resistance and heat conduction. Phonons play a crucial role in understanding the behavior of electrons in a crystal, especially in explaining why some materials conduct electricity well while others do not.

1.4 Mattheissen's rule

Matthiessen's rule is a principle that describes the total electrical resistivity of a crystalline material as the sum of its individual resistivities due to various scattering mechanisms.

In a crystalline material, the flow of electrical current is impeded by various scattering mechanisms that electrons encounter as they move through the lattice. These mechanisms can include impurities, defects, lattice vibrations (phonons), and electron-electron interactions. Matthiessen's rule states that the total electrical resistivity of the material is the sum of the resistivities associated with each of these scattering processes.

Mathematically, it can be expressed as:

$$\rho_{\text{total}} = \rho_{\text{impurities}} + \rho_{\text{defects}} + \rho_{\text{phonons}} + \rho_{\text{electron-electron interactions}} + \dots$$

Each term on the right side represents the contribution to the resistivity due to a specific scattering mechanism. By considering these individual contributions, Matthiessen's rule provides a way to estimate the total resistivity of a material.

But for our discussion, to make things simpler, we will say there are two kinds of contributions towards the total resistivity: a temperature-independent resistivity (ρ_0) due to impurities and a temperature-dependent resistivity (ρ_T) which is the sum of all other resistivities. So we can write Mattheissen's rule simply as:

$$\rho_{\text{total}} = \rho_0 + \rho_T$$

The variation of resistivity with temperature for pure and impure metals is as shown in figure 1.1.

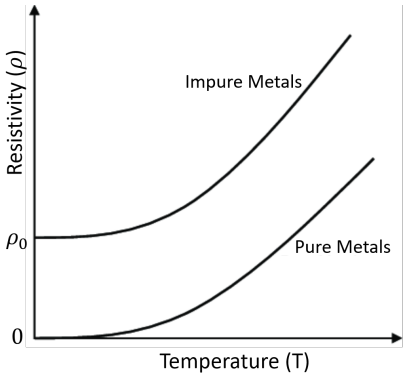


Figure 1.1: Mattheissen's rule

1.5 Quantum Free Electron Theory

The partial failure of classical free electron theory led Somerfield to propose the quantum free electron theory. He retained some of the assumptions of classical theory and introduced a few new ones.

1.5.1 Assumptions of Quantum Free Electron Theory

- The valence electrons are free to move about inside the metal. But they cannot escape from the metal due to the potential barrier at the surface.
- The force of attraction between the free electrons and the ionic core and the force of repulsion between electrons is negligible.
- The potential inside the metal is constant.
- **The energies of electrons are quantized.**
- **The allowed discrete energy levels obey Pauli's exclusion principle, according to which no two electrons in an energy level can have the same set of quantum numbers.**

What is meant by 'energy is quantized'?
Before the advent of quantum mechanics, the prevailing belief was that the energy of any given system could span an entire continuum from 0 to ∞ (which means **any** possible real number). However, subsequent observations revealed that **discrete** (non-continuous or specific) values of energy were characteristic of certain microscopic systems. For instance, the energy states of a hydrogen atom, the conduction and valence bands in solids, and the energy levels of a particle confined to a one-dimensional box (refer module 2) all exemplify this phenomenon. This concept of non-continuous, distinct energy values led to the term "quantum" (derived from the Latin word for quantity) mechanics. Importantly, it's worth noting that quantization doesn't solely apply to energy; it extends to other physical attributes such as angular momentum and particle "spin" (spin is a property of electrons just like mass and charge, but don't think of it as physical spinning).
So in the context of quantum free electron theory, we make the assumption that electrons within a metal do not possess an arbitrary range of energy values; instead, they are restricted to specific, discrete energy values. This gives rise to the idea of a energy level diagram, where along a vertical y-axis, we depict horizontal lines (levels) stacked one above the other to represent distinct energy values accessible to electrons within the metal. As you ascend on this diagram, the energy values increase. On these levels, we place dots to symbolize electrons (and we say the level is occupied by electron(s)). Importantly, there is a limit to the number of dots on each level, as dictated by Pauli's exclusion principle: a maximum of two electrons, each with opposite spin, can occupy a single energy level. Consequently, as you progress from the bottom, you can place a maximum of two dots on each energy level until all electrons are have been accounted for. At temperatures above 0 K, some energy levels may lack electrons or may contain only a single electron.

1.5.2 Density of States

Density of states $g(E)$ is defined as the number of *available* energy states (or levels) per unit energy range per unit volume, about a given energy E. From quantum mechanical considerations, it can be shown that for a metal

$$g(E)\mathrm{d}E = \frac{8\sqrt{2}\pi m^{3/2}}{h^3}E^{1/2}\mathrm{d}E$$

where h is Planck's constant and m is mass of electron.

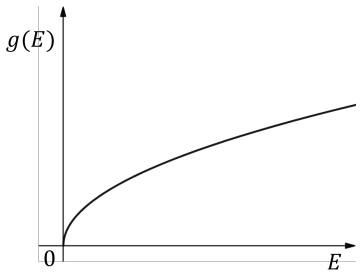


Figure 1.2: $g(E)$ vs. E

In simple words, the density of states tells us how many energy levels are available in a small energy-region $\mathrm{d}E$ around energy E. A plot of $g(E)$ versus E is as shown in figure 1.2, which we see is a parabola. The above formula says that if we have a fixed size of energy-region $\mathrm{d}E$ and we move to higher energy values, then the number of energy levels around those energy values increases. Note that this only tells us how many energy levels are available and not how many are occupied by electrons.

1.5.3 Fermi Energy

The highest energy that an electron can possess at 0 K in a metal is called Fermi energy and is denoted by E_F. The corresponding energy level is called the Fermi level (or Fermi energy level). At 0 K, all the energy levels above Fermi level are empty and all those below the Fermi level are completely filled, which will be shown later.

1.6 Fermi-Dirac Statistics

At temperatures above 0 K, one cannot tell with certainty whether a particular energy level will be occupied by an electron or not. Consequently, we need to assign probabilities to each energy level indicating the likelihood of it being occupied. For example, under thermal equilibrium, the free electrons occupy various energy levels in accordance with a statistical rule known as Fermi-Dirac statistics. Fermi-Dirac statistics enables the evaluation of probability of finding electron in energy levels over a certain range of energy values. The evaluation is done with the help of a quantity called Fermi factor $f(E)$ given by

$$f(E) = \frac{1}{1 + e^{\frac{E-E_F}{k_B T}}} \tag{1.1}$$

Fermi factor is the probability of occupation of a given energy state for a material at thermal equilibrium.

1.6.1 Dependence of Fermi Factor on Temperature and Energy

Since most measurements are made at room temperature, we must consider the effect of temperature on the electron gas. So it is important to know the variation of Fermi-Dirac distribution function with respect to temperature.

Case 1: $f(E)$ for $E < E_F$ at $T = 0$ K

When $E < E_F$ and $T = 0$ K, $\dfrac{E - E_F}{k_B T}$ becomes $-\infty$. Therefore (1.1) becomes

$$f(E < E_F)\Big|_{T=0\,\mathrm{K}} = \frac{1}{1 + e^{-\infty}} = 1$$

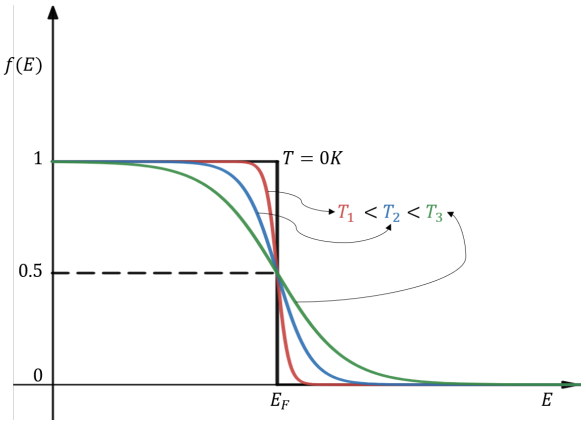


Figure 1.3: Variation of Fermi factor with energy for different temperatures

because $e^{-\infty} = 0$.

This means that at $T = 0\,\text{K}$, all the energy levels below the Fermi level are fully occupied by electrons, i.e., there is 100% probability of finding an electron below the Fermi energy level at absolute zero temperature.

Case 2: $f(E)$ for $E > E_F$ at $T = 0\,\text{K}$

When $E > E_F$ and $T = 0\,\text{K}$, $\dfrac{E - E_F}{k_B T}$ becomes ∞. Therefore (1.1) becomes

$$f(E > E_F)\Big|_{T=0\,\text{K}} = \frac{1}{1 + e^{\infty}} = 0$$

because $e^{\infty} = \infty$.

This means that at $T = 0\,\text{K}$, none of the energy levels above the Fermi level are occupied by electrons, i.e., there is 0% probability of finding an electron above the Fermi energy level at absolute zero temperature.

Case 3: $f(E)$ for $E = E_F$ at $T > 0\,\text{K}$

When $E = E_F$ and $T > 0\,\text{K}$, $\dfrac{E - E_F}{k_B T}$ becomes 0. Therefore (1.1) becomes

$$f(E = E_F)\Big|_{T>0\,\text{K}} = \frac{1}{1 + e^{0}} = \frac{1}{2}$$

because $e^{0} = 1$.

This means that at $T > 0\,\text{K}$, there is 50% probability of finding an electron at the Fermi energy level.

Figure 1.3 shows the variation of the Fermi factor with energy for different temperatures.

1.6.2 Worked Examples/Numericals

Q 1. Calculate the probability of occupation of an energy level 0.2 eV above fermi level at temperature 27 °C.

Given:

$$E = E_F + 0.2\,\text{eV}$$
$$E - E_F = 0.2eV = 0.2 \times 1.6 \times 10^{-19}$$
$$E - E_F = 3.2 \times 10^{-20}\,\text{J}$$

$$T = 27\,^\circ\text{C} = 27 + 300 = 300\,\text{K}$$

To find: $f(E) = ?$
We know that,

$$f(E) = \frac{1}{1 + \text{e}^{\frac{E-E_F}{k_B T}}} = \frac{1}{1 + \text{e}^{\frac{3.2\times10^{-20}}{1.38\times10^{-23}\times300}}} = 4.395 \times 10^{-4}$$

$$\boxed{f(E) = 4.395 \times 10^{-4}}$$

Q 2. Find the temperature at which there is 1% probability that a state with an energy 0.5 eV above Fermi energy is occupied.

Given:

$$f(E) = 1\% = 0.01$$

$$E = E_F + 0.5\,\text{eV}$$
$$E - E_F = 0.5eV = 0.5 \times 1.6 \times 10^{-19}$$
$$E - E_F = 8 \times 10^{-20}\,\text{J}$$

To find: $T = ?$
We know that,

$$f(E) = \frac{1}{1 + \text{e}^{\frac{E-E_F}{k_B T}}}$$

Rearranging the terms, we get,

$$T = \frac{E - E_F}{k_B \ln\left(\frac{1}{f(E)} - 1\right)} = \frac{8 \times 10^{-20}}{1.38 \times 10^{-23} \times \ln\left(\frac{1}{0.01} - 1\right)} = 1262\,\text{K}$$

$$\boxed{T = 1262\,\text{K}}$$

Q 3. The Fermi level in potassium is 2.1 eV. What is the energy of the energy level for which the probability of occupation at 300 K is 0.98?

Given:

$$E_F = 2.1eV = 2.1 \times 1.6 \times 10^{-19} = 3.36 \times 10^{-19}\,\text{J}$$

$$T = 300\,\text{K}$$

$$f(E) = 0.98$$

To find: $E = ?$

We know that,

$$f(E) = \frac{1}{1 + \mathrm{e}^{\frac{E-E_F}{k_B T}}}$$

Rearranging the terms, we get,

$$\begin{aligned} E &= E_F + k_B T \ln\left(\frac{1}{f(E)} - 1\right) \\ &= 3.36 \times 10^{-19} + 1.38 \times 10^{-23} \times 300 \times \ln\left(\frac{1}{0.98} - 1\right) \\ E &= 3.199 \times 10^{-19}\,\mathrm{J} \end{aligned}$$

In eV,

$$\boxed{E = 1.999\,\mathrm{eV}}$$

1.7 Model and Previous Year Questions

Q 1. Enumerate the failures of classical free electron theory and discuss the assumptions of quantum free electron theory of metals.

Q 2. Define Fermi energy level. Discuss various energy states by the electrons at $T = 0\,\mathrm{K}$ and $T > 0\,\mathrm{K}$ on the basis of fermi factor.

Q 3. Define Fermi factor. Discuss the variation of Fermi factor with temperature and energy.

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Chapter 2

Superconductivity

Syllabus
Superconductivity
Introduction to Super Conductors, Temperature dependence of resistivity, Meissner Effect, Critical Field, Temperature dependence of Critical field, Types of Super Conductors, BCS theory (Qualitative), Quantum Tunneling, High Temperature superconductivity, Josephson Junctions (Qualitative), DC and RF SQUIDs (Qualitative), Applications in Quantum Computing : Charge, Phase and Flux qubits. Numerical Problems.

2.1 Introduction

The story of superconductivity begins with Heike Kamerlingh Onnes, a Dutch physicist, who was conducting experiments at the University of Leiden during 1911. He was particularly interested in the behavior of materials at extremely low temperatures, using a specialized cryogenic apparatus to achieve temperatures very close to absolute zero ($-273.15\,^{\circ}\mathrm{C}$ or $0\,\mathrm{K}$).

Kamerlingh Onnes focused his attention on mercury, a common element, which he cooled to near-absolute-zero temperatures. His goal was to study how the electrical resistance of materials changes as they approach these incredibly low temperatures. To his amazement, he observed something extraordinary: as the temperature of the mercury dropped below a critical point, its electrical resistance suddenly and inexplicably vanished. This meant that electricity could flow through the material without any loss of energy due to resistance.

This phenomenon was so surprising that Kamerlingh Onnes initially doubted the accuracy of his own measurements. He meticulously repeated his experiments and found the same result every time. This marked the first documented observation of superconductivity, a term that would later be coined to describe materials that exhibit this remarkable property. Onnes' original graph depicting R vs. T for mercury (Hg) can be seen in figure 2.1.

He published his findings in 1911, describing the sudden drop in electrical resistance as a "zero resistance" state. This groundbreaking discovery earned him the Nobel Prize in Physics in 1913. Kamerlingh Onnes and his colleagues continued to investigate superconductivity, exploring other materials that exhibited this behavior and laying the foundation for future research into the underlying mechanisms.

Kamerlingh Onnes' discovery of superconductivity not only expanded our understanding of the behavior of matter at low temperatures but also sparked intense interest and research into the unique properties of superconducting materials. It set the stage for subsequent discoveries, the development of theoretical explanations like the BCS theory, and eventually,

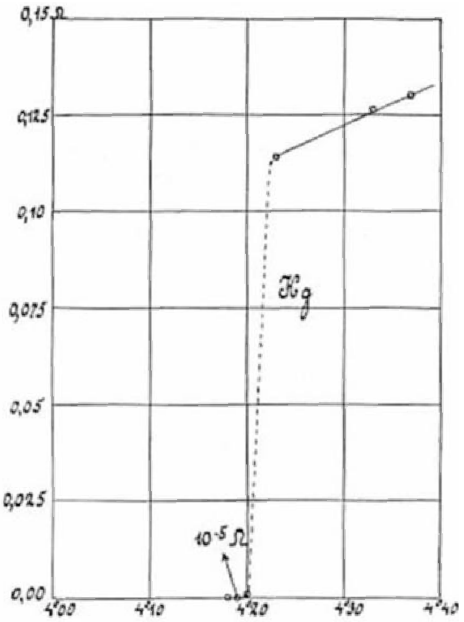


Figure 2.1: Onnes' original graph with
resistance on y-axis and temperature on x-axis

the practical applications of superconductors in a wide range of technologies.

2.1.1 Definition

Thus superconductivity can be defined as follows: *Superconductivity is the phenomenon in which the resistance of certain materials abruptly drops to zero below a certain temperature.* And superconductors are materials which exhibit the phenomenon of superconductivity.

2.1.2 Critical Temperature

The temperature below which materials exhibit superconducting property is called critical temperature and is denoted by T_c. Critical temperature is different for different materials.

Above the critical temperature, the material is said to be in normal state where it offers resistance for the flow of electric current. Below the critical temperature, the material is said to be in superconducting state where it exhibits superconducting behavior. Thus T_c is also called as transition temperature.

2.1.3 Variation of resistivity with temperature

The variation of resistivity of superconductors with temperature is as shown in figure 2.2. This figure also shows the variation of pure and impure metals, which was seen in figure 1.1, for comparison.

2.2 Meissner's Effect

Walther Meissner, a German physicist, and Robert Ochsenfeld were conducting experiments with superconductors at the University of Tübingen. They were particularly interested in the behavior of superconducting materials in the presence of magnetic fields. Up to that

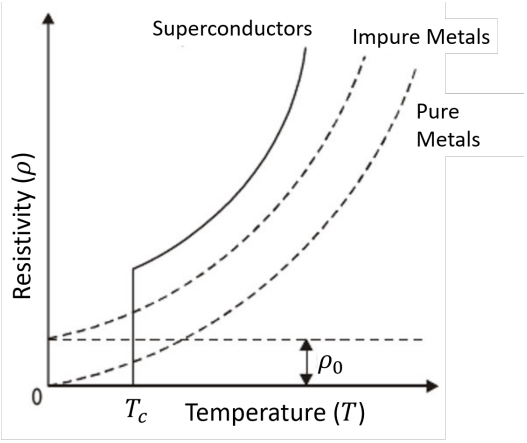


Figure 2.2: Variation of resistivity with temperature

point, researchers had already discovered the remarkable property of superconductivity - zero electrical resistance - but its interaction with magnetic fields remained unexplored.

During their experiments, Meissner and Ochsenfeld observed something extraordinary: when a superconducting material was cooled to its superconducting transition temperature and placed in a magnetic field, it exhibited an unexpected behavior. The magnetic field lines were not simply repelled; instead, they were expelled from the interior of the superconductor. This expulsion of magnetic flux created a unique effect in which the superconductor became a perfect diamagnet, meaning it experienced no magnetic field within its bulk.

The Meissner effect not only demonstrated that superconductors could expel magnetic fields but also highlighted that this phenomenon was a bulk property of superconducting materials, rather than a surface effect. It also laid the groundwork for the development of superconducting materials for powerful magnets used in technologies such as magnetic resonance imaging (MRI) and particle accelerators.

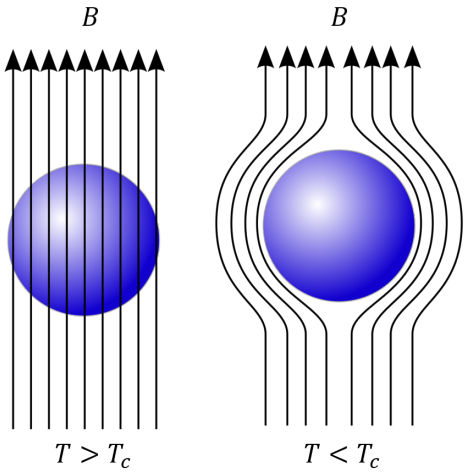


Figure 2.3: Meissner's Effect

Figure 2.3 shows an illustration of the Meissner's effect, with B representing the magnetic flux density and the sphere representing a solid superconductor. At temperatures exceeding the critical temperature ($T > T_c$), the material remains in its normal state, allowing magnetic field lines to permeate it. Conversely, when the temperature falls below the critical

temperature ($T < T_c$), the material transitions into a superconducting state, leading to the expulsion of magnetic field lines from the sphere's interior.

Now to prove that Meissner's effect leads to perfect diamagnetism, we will consider the following relation taken from the theory of electromagnetism:

$$B = \mu_0(M + H)$$

where B is the magnetic flux density, M is the magnetization, and H is the magnetic field strength.

Note:
Magnetic field strength (H) represents the intensity of magnetic field generated by an external current or a magnetizing force. Its unit is A/m (ampere per meter).
Magnetization (M) quantifies the extent to which a material becomes magnetized when subjected to an external magnetic field. Its unit is also A/m.
Magnetic flux density (B) represents the intensity of a magnetic field within a material or space. Its unit is T (tesla) or $\mathrm{Wb/m^2}$ (weber per meter-square).

Basically, when a material is exposed to an external magnetic field, the strength of that applied field is denoted as H. Due to the applied field, the material gets magnetized and produces its own magnetic field whose strength is denoted as M. The overall magnetic flux that is present in the material due to the external field and the material's own field is denoted as B.

Since the magnetic field lines are completely expelled from the bulk of the superconductor, we can say that the magnetic flux density is zero inside the superconductor, i.e., $B = 0$. So we have

$$\begin{aligned} 0 &= \mu_0(M + H) \\ M + H &= 0 \\ \frac{M}{H} &= -1 \\ \implies \chi_m &= -1 \end{aligned}$$

where $\chi_m = \frac{M}{H}$ is called the magnetic susceptibility of the material (magnetic susceptibility is a dimensionless quantity that indicates the degree of magnetization of a material in response to an applied magnetic field).

A material which has $\chi_m = -1$ is said to be perfectly diamagnetic. So our superconducting material exhibits perfect diamagnetism when it is below its critical temperature.

Origin of Meissner's effect

In the presence of an external magnetic field, superconducting materials adhere to Lenz's law. This law dictates that a changing magnetic field induces eddy currents within the superconductor. These eddy currents generate magnetic fields of their own, which counteract and repel the external magnetic field. Consequently, the superconductor effectively resists the intrusion of the magnetic field, preventing it from permeating the material. These unique currents are not your typical electric currents; they are referred to as "supercurrents." They are persistent flows of electrons that encounter no resistance. These supercurrents are exceptionally efficient, expelling any magnetic flux attempting to penetrate the material, resulting in the expulsion of all magnetic flux from within the material's bulk.

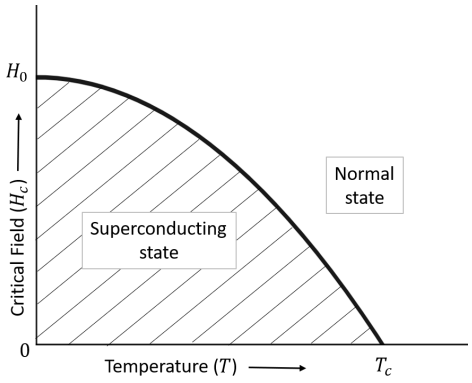


Figure 2.4: Critical field vs temperature

2.2.1 Critical Field

Can the superconductor keep expelling the magnetic field lines if we keep increasing the external field indefinitely? No. It is found that there is a limit to how much external field a superconductor can expel before turning into its normal state. This limit is called the critical field. Formally, *the critical field represents the maximum magnetic field strength that a superconducting material can tolerate before it transitions from a superconducting state to a normal (non-superconducting) state.* It is denoted as H_c.

However, it's important to note that it's feasible to recover the superconducting state even when the applied magnetic field surpasses the critical field. It has been observed that by further reducing the temperature, the superconducting state can be reinstated. Consequently, this implies an increase in the critical field value, necessitating a much higher magnetic field to disrupt superconductivity.

Then, according to the above discussion, we can say that the critical field is dependent on temperature, i.e., $H_c(T)$. This relation is mathematically written as follows:

$$H_c(T) = H_0 \left[1 - \frac{T^2}{T_c^2}\right]$$

where $H_c(T)$ is the critical field at some temperature T, H_0 is the critical field at 0 K temperature, T_c is the critical temperature of the material. T_c and H_0 are characteristic values for a specific material and are fixed. The above relation tells us that if we decrease the temperature of the material, the critical field (the amount of field required to break superconductivity) increases.

Figure 2.4 shows the variation of critical with temperature. Superconductivity is manifested in a material when its temperature and critical field value combination falls within the shaded area. Conversely, if the temperature and critical field value pair falls outside this region, the material will remain in its normal state.

The above expression for critical field can also be written in terms of the magnetic flux density as follows:

$$B_c(T) = B_0 \left[1 - \frac{T^2}{T_c^2}\right]$$

where $B_c(T)$ and B_0 represent the critical field at temperature T and 0 K respectively, with units of tesla.

Origin of critical field

If superconductors were devoid of any critical field limitation, it would imply an ability to expel an infinite amount of magnetic fields. However, this would necessitate the presence of infinite supercurrents, which in turn would require infinite energy - a physical impossibility. Alternatively, envision that as supercurrents increase indefinitely, electrons would eventually attain speeds surpassing the speed of light, violating fundamental physical constraints. Thus, superconductors possess a characteristic limit to the magnetic field they can sustain before forfeiting their superconducting property.

2.2.2 Worked Examples

Q 1. Lead has superconducting transition temperature of 7.26 K. If the initial field at 0 K is 50×10^3 A/m, calculate the critical field at 6 K.

Given:

$$T_c = 7.26\,\text{K}$$

$$H_0 = 50 \times 10^3\,\text{A/m} = 5 \times 10^4\,\text{A/m}$$

$$T = 6\,\text{K}$$

To find: H_c =?

We know that,

$$H_c = H_0\left[1 - \frac{T^2}{{T_c}^2}\right] = 5 \times 10^4 \times \left[1 - \frac{(6)^2}{(7.26)^2}\right] = 1.585 \times 10^4\,\text{A/m}$$

$$\boxed{H_c = 1.585 \times 10^4\,\text{A/m}}$$

Q 2. Superconducting tin has a critical magnetic field of 0.0217 T at 2 K. If the critical temperature for superconducting transition for tin is 3.7 K, find the critical magnetic field at 3 K.

Given:

Note that the critical field at 0 K is not given.

$$T_c = 3.7\,\text{K}$$

$$T_1 = 2\,\text{K}$$

$$B_{c_1} = 0.0217\,\text{T}$$

$$T_2 = 3\,\text{K}$$

To find: B_{c_2} =?

We know that,

$$B_c(T) = B_0\left[1 - \frac{T^2}{T_c^2}\right]$$

We need to write this for two different temperatures, i.e.,

$$B_{c_1}(T_1) = B_0\left[1 - \frac{T_1^2}{T_c^2}\right] \text{ and } B_{c_2}(T_2) = B_0\left[1 - \frac{T_2^2}{T_c^2}\right]$$

Taking the ratio of the second expression with the first, we get

$$B_{c_2} = B_{c_1} \frac{\left[1 - \frac{{T_2}^2}{{T_c}^2}\right]}{\left[1 - \frac{{T_1}^2}{{T_c}^2}\right]} = 0.0217 \times \frac{\left[1 - \frac{(3)^2}{(3.7)^2}\right]}{\left[1 - \frac{(2)^2}{(3.7)^2}\right]} = 0.0105\,\mathrm{T}$$

$$\boxed{B_{c_2} = 0.0105\,\mathrm{T}}$$

2.3 Types of Superconductors

Superconductors are classified into two types:

1. Type I superconductors
2. Type II superconductors

(Very creative names, I know.)

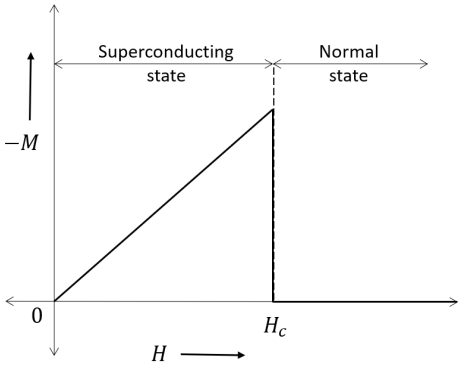


Figure 2.5: Type I Superconductor

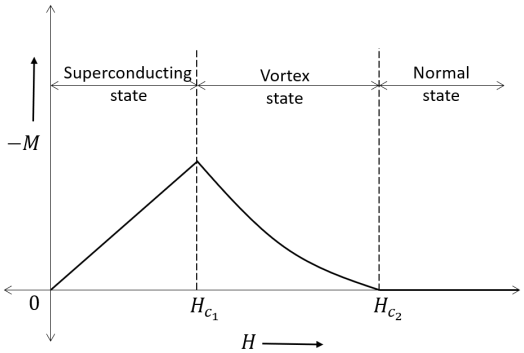


Figure 2.6: Type II Superconductor

2.3.1 Type I superconductors

Type I superconductors demonstrate complete Meissner's effect and are characterized by a single critical field value. The graph depicting negative magnetization versus magnetic field strength for these superconductors is illustrated in figure 2.5. With increasing field strength, the material exhibits an increasingly diamagnetic response until the field strength equals H_c. Beyond this threshold, the material permits flux lines to penetrate, leading to normal conductivity.

These superconductors possess an exceptionally low critical field value, rendering them vulnerable to high magnetic fields. Because magnetization abruptly vanishes upon reaching the critical field, they are often termed "soft superconductors" and hence have limited practical applications. Examples of Type I superconductors include Hg, Pb, and Zn.

2.3.2 Type II superconductors

Type II superconductors exhibit both complete and partial Meissner's effects, depending on the magnetic field, and are characterized by two distinct critical field values, denoted as H_{c_1} and H_{c_2}, with $H_{c_1} < H_{c_2}$. The graph illustrating the relationship between negative

magnetization and magnetic field strength for these superconductors is provided in figure 2.6.

As the field strength increases, the material increasingly displays a perfect diamagnetic response (superconducting state), progressing until the field strength equals H_{c_1}. Beyond this first critical field, the magnetization starts to gradually decrease as the magnetic field continues to rise (as opposed to the abrupt drop seen in Type I superconductors) until it reaches H_{c_2}. In this phase between H_{c_1} and H_{c_2}, also called the Vortex state, the material exhibits partial Meissner's effect, allowing limited magnetic flux penetration. Beyond the second critical field, the material permits the passage of all magnetic flux lines, returning to a state of normal conductivity.

Type II superconductors possess a notably higher second critical field value H_{c_2} compared to H_c of type I superconductors, enabling them to withstand substantial magnetic fields. Because magnetization diminishes *gradually* upon approaching the critical field rather than disappearing abruptly, they are often referred to as "hard superconductors" and find numerous practical applications such as in superconducting magnets used in MRIs and high-speed trains. Examples of Type II superconductors include NbTi, Nb_3Sn.

More on vortex state: When a Type II superconductor is subjected to an applied magnetic field that exceeds its first critical field (H_{c_1}), magnetic flux begins to penetrate the material in quantized units known as flux vortices. These vortices carry discrete amounts of magnetic flux and create a lattice-like arrangement within the superconducting material. Each vortex has a normal (non-superconducting) core where the superconductivity is locally suppressed (where flux is allowed to pass) and is surrounded by circulating superconducting currents; hence the name "vortex".

2.4 BCS Theory of Superconductivity

The Bardeen-Cooper-Schrieffer (BCS) theory is a microscopic theory of superconductivity, which explains how certain materials can conduct electric current without any resistance when cooled below a certain temperature.

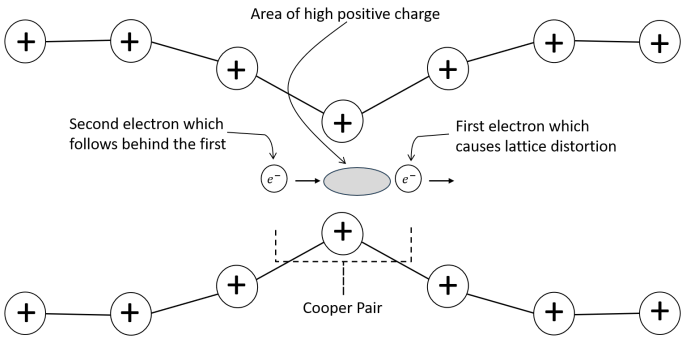


Figure 2.7: BCS Theory - Formation of Cooper Pair

The theory goes as follows: In a superconducting material, the atomic nuclei form a lattice structure. When an electron moves through this lattice, it causes a slight displacement of the nearby positive nuclei due to its negative charge. This displacement propagates through the lattice as a quantized vibration, known as a phonon. Refer to figure 2.7.

Now, consider another electron moving through the lattice. This electron can interact with the phonon, effectively being attracted to the region of positive charge created by the

displacement of the nuclei. This results in an effective attractive interaction between the two electrons, which is counterintuitive because electrons, being negatively charged, should repel each other due to their like charges. This kind of interaction between two electrons mediated by a phonon is also called a electron-phonon-electron interaction.

This attractive interaction leads to the formation of special pairs of electrons known as **Cooper pairs**, named after Leon Cooper. These pairs are formed from electrons with opposite spins and momenta. So the pair as a whole has a total spin of zero. These pairs are unusual because they act like a single entity, moving through the lattice together without scattering off impurities or lattice vibrations (phonons), which is what normally causes electrical resistance. As a result, they can flow without any loss of energy or dissipation, leading to superconductivity.

Origin of critical temperature

It's important to note that this pairing mechanism and the resulting superconductivity only occur at temperatures below the critical temperature. Above this temperature, thermal vibrations of the lattice are too strong and disrupt the Cooper pairs, causing the material to lose its superconducting properties and behave as a normal conductor.

The following is additional information provided for reference only.
The fact that Cooper pairs have a total spin of zero is significant because particles with integer spins (like 0) are classified as bosons, while particles with half-integer spins (like 1/2) are fermions.
Electrons, being fermions, obey the Pauli's Exclusion Principle, which states that no two fermions can occupy the same quantum state simultaneously. However, when electrons pair up to form Cooper pairs, they effectively behave as bosons and do not obey the Pauli Exclusion Principle. This means that many Cooper pairs can exist in the same quantum state.
When a large number of Cooper pairs form in a superconductor below its critical temperature, they create something called a Bose-Einstein condensate. In this state, the Cooper pairs move coherently as if they were one "superparticle", rather than as individual pairs. This coherence allows them to move through the lattice without scattering off impurities or lattice vibrations (phonons), which is what normally causes electrical resistance. The formation of this Bose-Einstein condensate of Cooper pairs and their coherent motion is what enables superconductivity.

2.5 High-temperature Superconductors

High-temperature superconductors are materials that exhibit superconducting properties at temperatures much higher than traditional superconductors. While "high temperature" in this context still refers to temperatures that are quite cold by everyday standards, these materials are exciting because they operate above the boiling point of liquid nitrogen ($-196\,^{\circ}\mathrm{C}$ or $77\,\mathrm{K}$), making them more practical for real-world applications. (The critical temperatures of traditional superconductors are far below the boiling point of liquid nitrogen and require cooling with liquid helium, which is more expensive and harder to handle than liquid nitrogen.)

The first high-temperature superconductor was discovered in 1986 by IBM researchers Georg Bednorz and K. Alex Müller, who found that a lanthanum-based cuprate perovskite material exhibited superconductivity at a temperature of $35\,\mathrm{K}$. They were awarded the

Nobel Prize in Physics the following year for their discovery. Since then, many other high-temperature superconductors have been discovered. Most of these are cuprates, which are compounds made up of copper and oxygen, along with other elements. Some popular examples include:

- Yttrium barium copper oxide (YBCO): This is one of the most well-known high-temperature superconductors. It was the first material discovered to have a critical temperature above 77K, which is significant because it means it can be cooled with liquid nitrogen, which is cheaper and more easily available than other coolants like liquid helium. The highest critical temperature for YBCO is around 93K. YBCO's chemical formula is written as $YBa_2Cu_3O_7$ and hence it is also called the 1-2-3 compound.
- Bismuth strontium calcium copper oxide (BSCCO): Also known as BISCO, this material has a critical temperature of up to 108K.
- Mercury barium calcium copper oxide (HBCCO): With a critical temperature as high as 138K under pressure, this is one of the highest high-temperature superconductors known.
- Iron-based superconductors: Discovered in 2008, these are a new family of high temperature superconductors. They have a different structure from the cuprates, and their highest critical temperature is around 56K.

It's important to note that while these materials can conduct electricity without resistance at higher temperatures than traditional superconductors, we still don't fully understand how they work. The BCS theory that explains low-temperature superconductivity does not adequately describe high-temperature superconductivity, and developing a comprehensive theory of high-temperature superconductivity is one of the major unsolved problems in theoretical physics.

2.6 Quantum Tunnelling

Quantum tunnelling is a fascinating phenomenon that arises from the principles of quantum mechanics. It allows particles to do things that would seem impossible according to classical physics.

In classical physics, a particle cannot cross a potential energy barrier (analogous to a hill) if its energy is less than the height of the barrier. It would be like saying a ball could roll up and over a hill even if it doesn't have enough speed (and therefore energy) to reach the top. (Let me clarify something here to avoid confusion. The barrier that we talk about here is not a physical barrier; it's an *energy* barrier. The higher the barrier, the more energy it has.)

However, in quantum mechanics, particles such as electrons are described by wavefunctions, which are mathematical functions that describe the probability distribution of a particle's position. These wavefunctions can extend beyond potential barriers, even when the particle's energy is less than the height of the barrier. This means there's a non-zero probability that the particle can be found on the other side of the barrier, even though it seemingly doesn't have enough energy to get there. This is quantum tunnelling - the particle has effectively 'tunnelled' through the barrier. Refer to figure 2.8.

Quantum tunnelling plays a crucial role in many physical phenomena and technological applications. For example, it's responsible for nuclear fusion in stars, radioactive decay

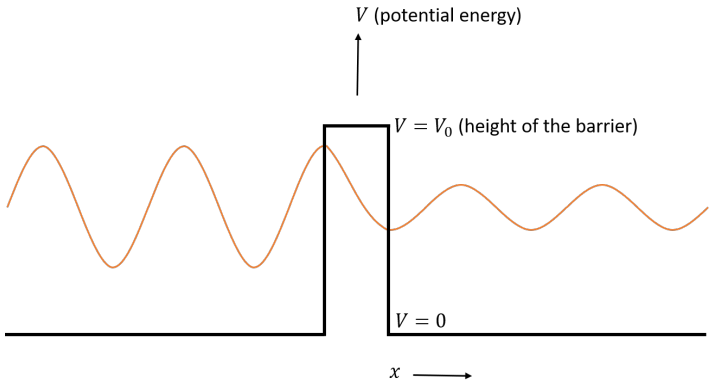
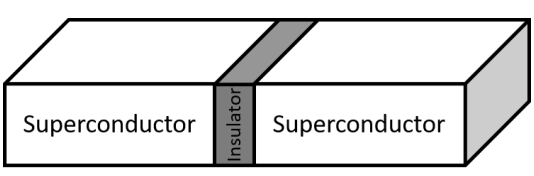


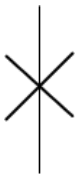
Figure 2.8: Quantum Tunnelling

of atomic nuclei, and the operation of tunnel diodes and scanning tunnelling microscopes. Another example of quantum tunnelling can be found in something called the Josephson junction which we will study next.

2.7 Josephson Junction



(a) Construction



(b) Circuit symbol of JJ

Figure 2.9: Josephson Junction

Josephson Junction is a device named after British physicist Brian Josephson, who predicted its behavior in 1962.

The junction is made up of two superconducting materials separated by a very thin barrier. Refer figure 2.9a. This barrier can be an insulator, a non-superconducting metal, or a physical constriction that weakens the superconductivity at the point of contact. The barrier is thin enough that electrons, or Cooper-paired electrons in the superconducting state, can tunnel through the barrier at an appreciable rate. The circuit symbol for a Josephson junction (JJ) is as shown in figure 2.9b.

This phenomenon of tunnelling of electrons through the junction is called the Josephson effect, and it is of two types: DC Josephson effect and AC Josephson effect.

2.7.1 DC Josephson Effect

The DC Josephson Effect refers to the flow of a supercurrent across a Josephson junction in the absence of any external voltage. This supercurrent is a direct current (DC) that flows continuously without any voltage applied. The supercurrent arises due to the tunneling of Cooper pairs across the thin barrier separating the two superconductors.

The Cooper pairs in a superconductor behave as a unified entity (as was discussed before), which means they share the same wavefunction, thus allowing us to describe the behavior of

electrons in a superconductor with a single wavefunction. This wavefunction is characterized by a property known as the phase.

In the context of a Josephson junction, the insulating barrier plays a crucial role in ensuring that the wavefunctions of the two superconductors have differing phases. This results in a phase difference across the junction. Brian Josephson demonstrated that this phase difference is responsible for the generation of a supercurrent within the junction, as described by the formula:

$$I_S = I_C \sin \varphi_0$$

where I_S represents the supercurrent, I_C is the critical current (material-dependent characteristic value), and φ_0 signifies the phase difference across the junction.

2.7.2 AC Josephson Effect

The AC Josephson Effect, on the other hand, occurs when there is a fixed voltage across the junction. Refer figure 2.10. In this case, the phase difference across the junction varies linearly with time. This leads to an alternating current (AC) across the junction, with a frequency directly proportional to the applied voltage.

The formula for the supercurrent is:

$$I_S = I_C \sin(\varphi_0 + \Delta\varphi)$$

where $\Delta\varphi$ is the additional phase introduced due to the provided voltage V and is given by the formula $\Delta\varphi = 2eVt/\hbar$, and the other terms have the same meaning as before.

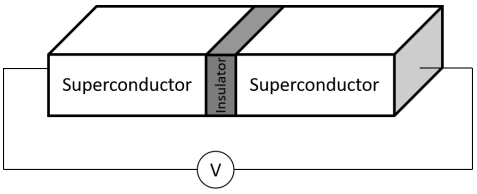


Figure 2.10: AC Josephson Effect

Both the DC and AC Josephson effects are examples of macroscopic quantum phenomena, where the effects of quantum mechanics are observable at ordinary, rather than atomic, scales. They have many practical applications in quantum-mechanical circuits, such as SQUIDs (Superconducting Quantum Interference Devices), superconducting qubits, etc.

2.8 SQUID

A **S**uperconducting **Qu**antum **I**nterference **D**evice (SQUID) is a highly sensitive magnetometer used to measure extremely weak magnetic fields. It consists of a superconducting loop interrupted by one or two Josephson junctions.

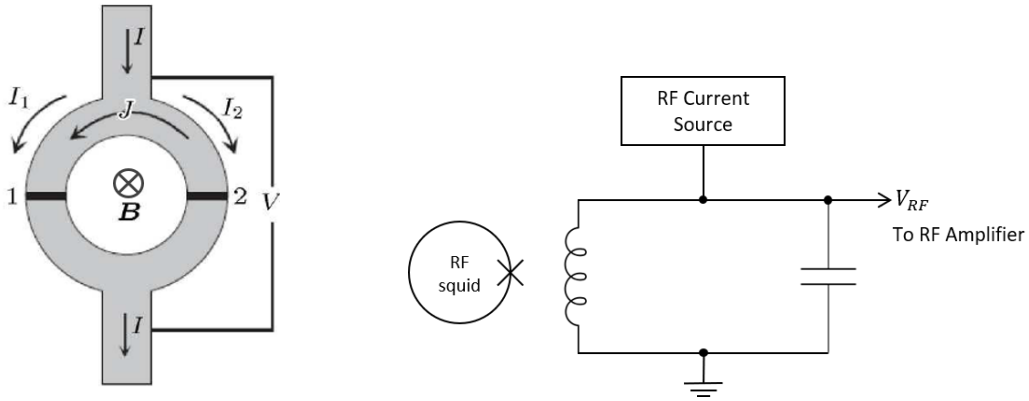
There are two types of SQUIDs: DC SQUID and RF SQUID.

2.8.1 Flux Quantization

Before delving into the SQUID types, let's explore flux quantization, which is a key idea behind the working of a SQUID.

Magnetic flux, denoted by Φ and measured in webers (Wb), signifies the strength of magentic field lines. It has been experimentally observed that in a superconducting *loop* (think of a bulk superconductor with a "hole"), magnetic flux that threads the loop can not be arbitrary; it's quantized. The flux Φ inside the loop must be an integral multiple of the flux quantum Φ_0, represented as:

$$\Phi = n\Phi_0$$



(a) DC SQUID (b) RF SQUID

Figure 2.11: Types of SQUIDs

where n is a positive integer, and Φ_0 is defined as

$$\Phi_0 = \frac{h}{2e} = 2.07 \times 10^{-15}\,\text{Wb}$$

2.8.2 DC SQUID

The DC SQUID consists of a superconducting ring with two parallel Josephson junctions, preferably with similar properties. Refer to figure 2.11a.

In the absence of an external magnetic field, the constant biasing current I evenly divides into two branches, I_1 and I_2, each equal to $I/2$. When a small external magnetic field B is applied to the superconducting loop, a screening current J circulates the loop, generating a magnetic field that opposes the external flux, following Lenz's law. The induced current aligns with I in one branch and opposes I in the other branch. Consequently, the total current increases in one branch $(I_1 + J)$ and decreases in the other $(I_2 - J)$. When the current in either branch surpasses the critical current I_C of the Josephson junction, a voltage emerges across the junction.

Now, suppose the external flux further increases until it exceeds $\Phi_0/2$, half of the magnetic flux quantum. The SQUID, adhering to energy preferences, shifts from screening the flux to increasing it to Φ_0. Consequently, the current now opposes the difference between the admitted flux (Φ_0) and the external field just over $\Phi_0/2$. As the external field increases, the current diminishes, reaching zero at exactly Φ_0, and then switches direction with further increases in the external field. The current periodically changes direction each time the flux increases by an additional half-integer multiple of Φ_0. The result is an oscillating current in the SQUID loop that corresponds to the applied magnetic field.

The voltage across the junctions also oscillates with the applied magnetic field. This oscillation can be detected and used to measure the magnetic field with high precision. The voltage oscillation is periodic with the flux quantum, and the period of the oscillation gives the value of the flux quantum.

2.8.3 RF SQUID

The RF SQUID consists of a superconducting loop interrupted by a single Josephson junction. Refer to figure 2.11b. The flux is coupled into the loop through an input coil (not shown in

diagram) and an RF coil. The RF coil is part of a high-Q resonant circuit that reads out changes in current due to induced flux in the SQUID loop. The tuned circuit is driven by a constant RF oscillator which is weakly coupled to the loop.

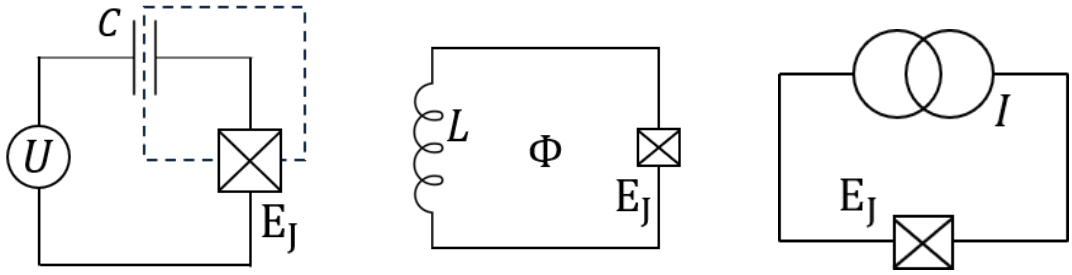
When a magnetic field is applied to the RF SQUID, it induces a change in the magnetic flux through the loop. This change in flux alters the phase difference across the Josephson junction, which in turn changes the supercurrent flowing through the junction. The change in supercurrent is then detected by the RF coil, which is part of a resonant circuit. The resonant frequency of this circuit changes with the supercurrent, allowing the change in magnetic flux to be measured.

The RF SQUID is less sensitive compared to the DC SQUID but is cheaper and easier to manufacture in smaller quantities.

2.9 Applications of superconductors in quantum computing

Superconducting qubits are a cornerstone of quantum computing, leveraging the unique properties of superconductors to store and process quantum information. They are implemented in superconducting circuits that can exist in two states, representing the binary 0 and 1, similar to classical bits. However, unlike classical bits, these qubits can exist in a superposition of states, enabling them to perform multiple calculations simultaneously.

There are three primary types of superconducting qubits: charge qubit, flux qubit, and phase qubit.



(a) Charge qubit (b) Flux qubit (c) Phase qubit

Figure 2.12: Superconducting qubits

2.9.1 Charge qubit

Also known as Cooper-pair boxes, charge qubits are qubits whose basis states are charge states, representing the presence or absence of excess Cooper pairs in a superconducting island (shown as dotted region in figure 2.12a). The state of the qubit is determined by the number of Cooper pairs that have tunneled across a Josephson junction.

The operation of charge qubits involves applying a microwave pulse to the gate of the Cooper-pair box. This causes the qubit to oscillate between its two states. By carefully controlling the amplitude and duration of the pulse, it's possible to perform arbitrary single-qubit operations. Despite their sensitivity to charge noise, charge qubits have been successfully used in many quantum computing experiments.

2.9.2 Flux qubit

Flux qubits, also known as persistent current qubits, are micrometer-sized loops of superconducting metal interrupted by a number of Josephson junctions. Refer to figure 2.12b. These

devices function as quantum bits, with the two lowest energy eigenstates of the loop being clockwise and counter-clockwise currents.

The state of a flux qubit is determined by the direction of the current flowing around the loop. By applying a magnetic field to the loop, it is possible to control the state of the qubit. Flux qubits are less sensitive to charge noise than charge qubits, but they are more sensitive to flux noise.

2.9.3 Phase qubit

Refer to figure 2.12c. The state of the phase qubit is determined by the phase difference across the Josephson junction. Phase qubits use the two lowest energy levels in the local minimum of something called the washboard potential as computational states.

The operation of phase qubits involves applying a microwave pulse to the junction. This causes the qubit to oscillate between its two states. By carefully controlling the amplitude and duration of the pulse, it is possible to perform arbitrary single-qubit operations. They are less sensitive to charge noise compared to charge qubits.

2.10 Model and Previous Year Questions

Q 1. Explain DC and AC Josephson effects and mention any two applications ot superconductivity in quantum computing.

Q 2. Discuss the effect of temperature and impurity on electrical resistivity of conductors and hence explain for superconductors.

Q 3. Explain Meissner's effect and the variation of critical field with temperature.

Q 4. Define critical temperature and critical magnetic field. Explain briefly BCS theory of suerconductivity.

Q 5. Distinguish between Type - I and Type - II superconductors.

Q 6. Explain BCS theory of superconductivity.

Q 7. Give the qualitative explanation of RF SQUID with the help of a neat sketch.

2.11 Numericals

Q 1. The critical teniperature of Nb is 9.15 K. At zero kelvin, the critical field is 0.196 T. Calculate the critical field at 8 K.

Q 2. A superconducting tin has a critical temperature of 3.7 K at zero magnetic field and a critical field of 0.0306 T at 0 K. Find the critical field at 2 K.

Q 3. The superconducting transition temperature of Lead is 7.26 K. Calculate the initial field at 0 K given the critical field at 5 K as 33.644×10^3 A/m.

Q 4. Calculate the ratio of critical fields for a superconductor at 7 K and 5 K given the critical temperature 8 K.

Q 5. The critical field for niobium is 1×10^5 A/m at 8 K and 2×10^5 A/m at 0 K. Calculate the transition temperature of the element.